

GAVLE, Sweden

WMO: 024530

Lat: 60.7161N

Lon: 17.1607E

Elev: 16

StdP: 101.13

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -19.3	(c) -15.8	(d) -21.8	(e) 0.5	(f) -18.7	(g) -18.2	(h) 0.8	(i) -15.5	(j) 8.7	(k) 1.1	(l) 7.9	(m) -0.4	(n) 0.6	(o) 200	(p) 0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.6	(c) 25.9	(d) 18.1	(e) 24.0	(f) 17.0	(g) 22.2	(h) 16.3	(i) 19.2	(j) 23.7	(k) 18.1	(l) 22.2	(m) 17.2	(n) 20.8	(o) 3.3	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	12.7	20.8	16.6	11.8	19.8	15.6	11.1	18.9	54.8	23.6	51.3	22.4	48.3	20.9	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.6	6.7	5.9	-23.2	28.4	3.4	2.0	-25.7	29.8	-27.7	31.0	-29.6	32.1	-32.1	33.5
			-23.1	20.7	3.2	1.5	-25.4	21.8	-27.3	22.6	-29.1	23.4	-31.4	24.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-3.8	-3.1	-0.4	4.2	8.9	13.3	15.9	15.2	11.2	5.7	1.6	-1.9
	DBStd	8.03	5.62	5.37	4.71	3.54	3.67	3.05	2.64	2.75	3.11	3.64	3.96	5.33
	HDD10.0	2148	427	367	323	178	63	6	0	1	23	139	254	368
	HDD18.3	4659	686	601	581	425	292	154	84	103	214	391	503	626
	CDD10.0	548	0	0	0	3	31	103	184	162	59	7	0	0
	CDD18.3	19	0	0	0	0	1	2	10	6	1	0	0	0
	CDH23.3	228	0	0	0	0	11	34	115	67	2	0	0	0
	CDH26.7	27	0	0	0	0	0	3	17	7	0	0	0	0
(14) Wind	WSAvg	2.7	2.6	2.8	2.9	2.8	3.0	2.8	2.5	2.4	2.5	2.6	2.7	2.7
(15) Precipitation	PrecAvg	632	46	32	32	41	39	51	73	80	65	58	60	53
	PrecMax	884	120	98	74	100	116	187	166	253	214	169	144	142
	PrecMin	443	4	6	3	2	0	12	15	0	12	12	13	0
	PrecStd	119	25	19	17	26	28	34	40	53	41	38	31	34
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	8.3	13.2	19.5	24.9	26.8	28.8	27.6	22.9	16.2	11.7	9.7
		MCWB	4.9	5.3	7.3	11.1	15.2	17.5	19.5	18.4	16.4	12.6	9.9	8.0
	2%	DB	5.3	6.6	10.2	16.2	21.1	23.8	26.3	25.3	20.5	14.1	10.0	7.4
		MCWB	3.6	4.0	5.7	9.5	13.3	15.7	18.7	18.1	15.2	11.6	8.4	6.0
	5%	DB	4.3	5.1	8.1	13.3	18.6	21.8	24.3	23.1	18.5	12.7	8.4	6.0
		MCWB	2.9	3.2	4.4	8.1	12.0	15.1	17.7	17.2	14.3	10.8	7.2	4.7
	10%	DB	3.1	3.7	6.2	11.0	16.3	19.9	22.3	21.2	16.9	11.3	7.1	4.4
		MCWB	2.0	2.1	3.3	6.8	10.7	14.2	16.9	16.5	13.5	9.7	6.2	3.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	5.8	7.7	12.1	16.3	18.5	20.8	20.9	17.5	13.5	10.2	8.0
		MCDB	6.4	7.8	12.0	18.5	22.8	24.1	26.3	24.6	20.8	15.3	11.4	9.1
	2%	WB	4.1	4.5	6.2	10.0	14.1	17.0	19.5	19.2	16.0	12.2	8.6	6.1
		MCDB	5.0	6.1	9.5	15.2	19.3	21.6	24.6	22.9	19.1	13.6	9.7	7.1
	5%	WB	3.1	3.4	4.9	8.5	12.6	15.9	18.5	18.1	15.0	11.1	7.4	4.7
		MCDB	4.1	4.8	7.5	12.6	17.6	20.6	22.9	21.6	18.0	12.4	8.2	5.6
	10%	WB	2.1	2.2	3.5	7.1	11.2	14.8	17.4	17.1	14.0	9.9	6.3	3.4
		MCDB	3.1	3.5	5.9	10.4	15.6	18.9	21.4	20.3	16.3	11.1	7.1	4.3
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	6.7	8.1	9.2	9.9	9.3	9.6	10.1	9.1	6.8	4.6	5.1
		MCDBR	5.1	6.8	10.1	14.3	15.0	13.4	13.7	13.5	11.5	8.2	6.2	5.2
		MCWBR	4.6	5.1	6.6	8.5	8.1	6.8	7.0	7.0	6.9	6.1	5.4	4.5
	5% WB	MCDWR	4.9	6.3	9.5	13.0	12.9	11.6	11.9	11.1	10.3	7.5	5.8	4.8
		MCWBR	4.5	5.0	6.5	8.0	7.6	6.5	6.9	6.6	6.7	5.9	5.2	4.4
(40) Clear-Sky Solar Irradiance	taub		0.262	0.308	0.330	0.364	0.364	0.357	0.380	0.372	0.339	0.325	0.285	0.366
	taud		2.189	2.280	2.358	2.353	2.417	2.472	2.428	2.444	2.523	2.439	2.240	2.655
	Ebn at Noon		491	679	796	831	860	873	840	817	788	662	452	343
	Edh at Noon		38	65	84	101	102	99	101	92	72	56	37	26
(44) All-Sky Solar Radiation	RadAvg		0.27	0.90	2.39	3.86	4.93	5.49	5.11	3.97	2.57	1.17	0.35	0.16
	RadStd		0.04	0.16	0.28	0.37	0.51	0.39	0.52	0.40	0.33	0.19	0.08	0.02

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-100	N/A	N/A	N/A

Nomenclature: See separate page